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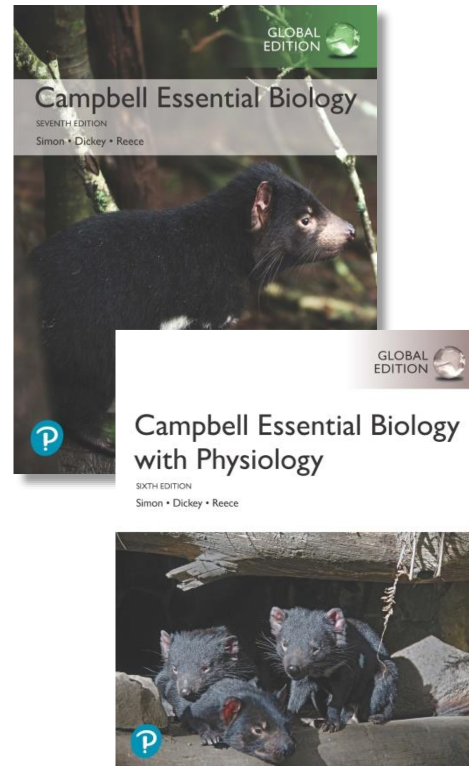
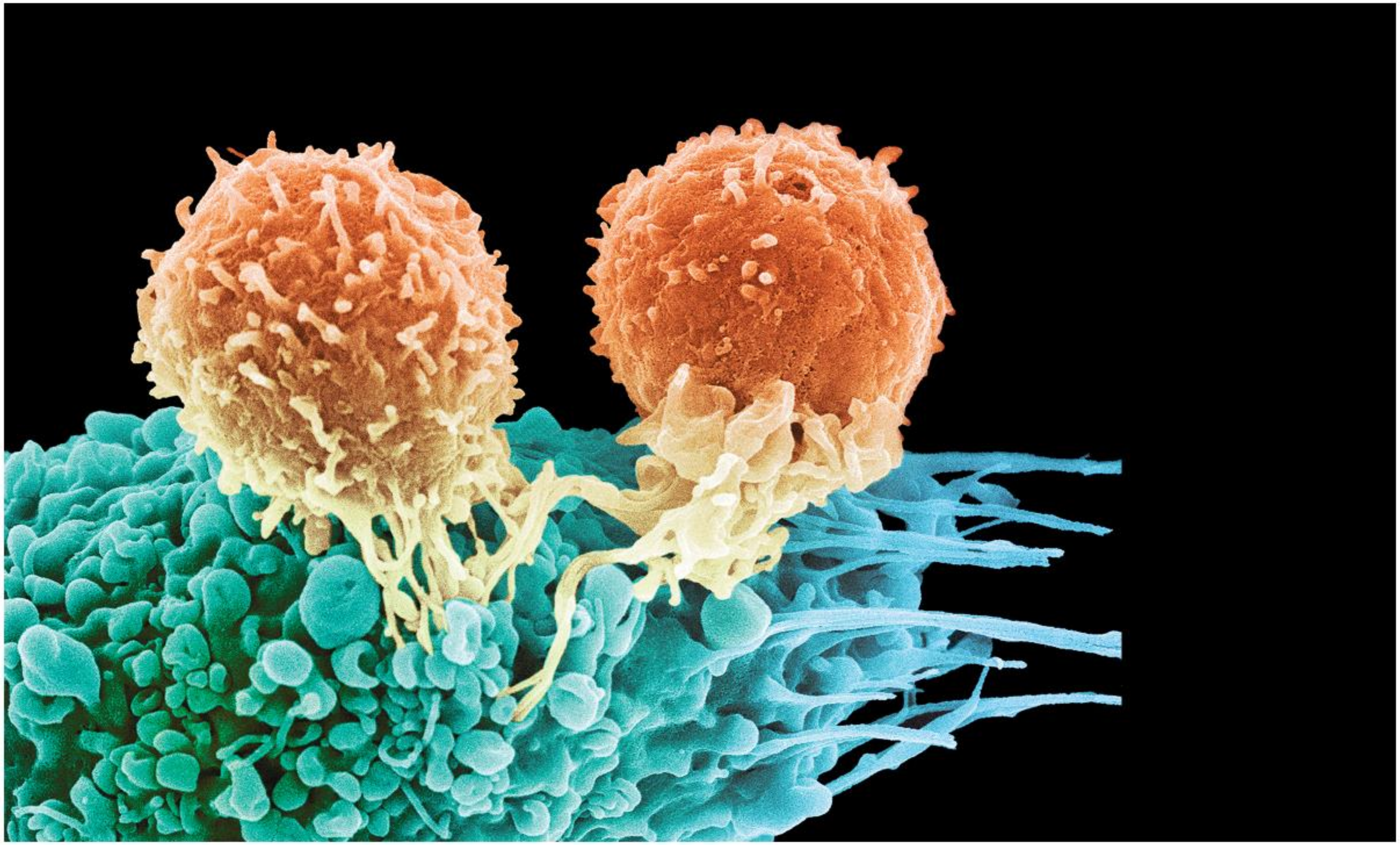


Chapter 24
The Body’s Defenses

PowerPoint® Lectures created by Edward J. Zalisko, Eric J. Simon, Jean L. Dickey, and Jane B. Reece

Pearson

Killers Lurk Within Your Body: The Sole Function Of Billions of Your Cells Is To Kill Other Cells.



Biology and Society: Herd Immunity (1 of 2)

- In April 2015, 147 people in seven states had been infected with measles, spread by a visitor to Disneyland in California.
  - Measles is caused by a highly contagious virus.
  - Just a few decades ago measles killed more than a million people each year, worldwide.
  - By 2000, widespread vaccination had virtually eliminated the disease in the United States.
  - But measles and other infectious diseases can return to communities that have low vaccination rates.

Measles: 홍역, RNA 바이러스 질환

Vaccinating a community

Biology and Society: Herd Immunity (2 of 2)

- The more people who are vaccinated against viral diseases, the safer it is for everyone.
  - This community protection, referred to as "herd immunity," occurs because the virus cannot take hold within the group.
  - Herd immunity is the rationale behind state-mandated vaccinations for children in public schools and national health campaigns reminding you to get your flu vaccine each year.
  - Research shows that herd immunity fails when just 5% of a community is unvaccinated.

Herd immunity: 집단 면역, 바이러스가 감염시킬 숙주를 찾지 못해 스스로 사멸하게 되는 현상

An Overview of the Immune System (1 of 3)

- The **immune system** is your body's defense against infectious disease.
- The human immune system is a collection of organs, tissues, and cells that together perform the vital function of safeguarding the body from a constant barrage of **pathogens**, disease-causing agents like viruses and bacteria.

An Overview of the Immune System (2 of 3)

- The protection provided by your immune system consists of two parts.
    1. **Innate immunity** doesn't change much from the time you are born, and its components attack pathogens indiscriminately.
    2. **Adaptive immunity** continually develops over your lifetime as it encounters and attacks specific pathogens. With its continual development, adaptive immunity ensures you will have better protection from a specific pathogen each time you encounter it.

Adaptive immunity: 적응 면역, 후천 면역

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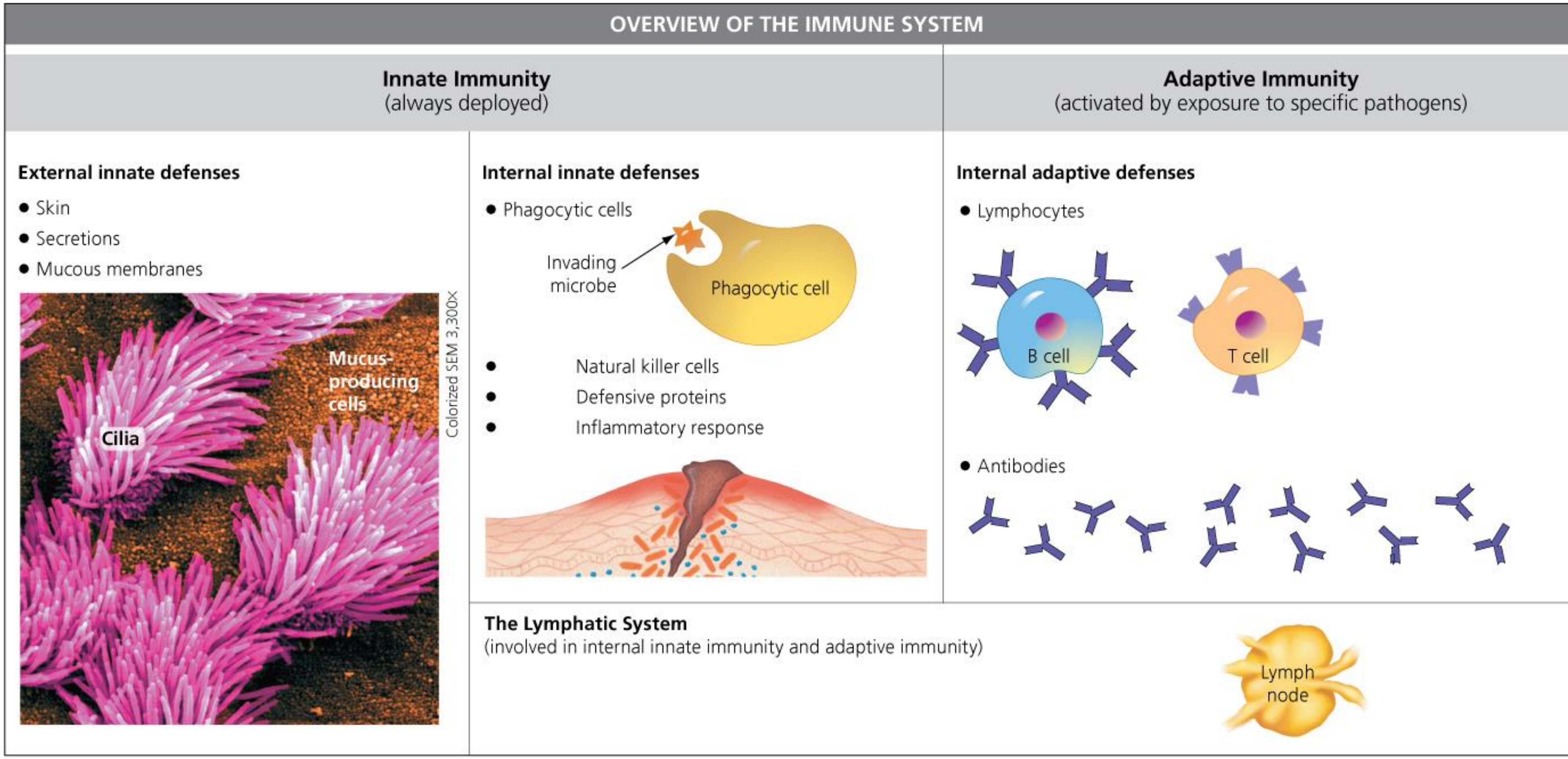
Overview of the body's defenses



An Overview of the Immune System (3 of 3)

- Scientists suspect that innate immunity evolved before adaptive immunity because innate immunity is found in nearly all animals, but only vertebrates have adaptive immunity.
- The **lymphatic system** is a network of vessels, tissues, and organs that lies at the intersection of the innate and adaptive responses.

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Innate Immunity

- Innate immunity protects the body when a pathogen first attempts to infect the body.
- This protection is accomplished with two lines of defense.
  1. External innate defenses prevent pathogens from getting deep inside the body.
  2. Internal innate defenses confront pathogens that make it past external defenses using immune cells and defensive proteins.

External Innate Defenses (1 of 2)

- External innate defenses form the frontline of your immune system because they *prevent* infection, as opposed to your body's other defenses, which fight an infection *after* it occurs.
- Some barriers of external innate defenses block or filter out pathogens.
  - Intact skin forms a tough outer layer that most bacteria and viruses cannot penetrate.
  - Organ systems that are open to the external environment are lined with cells that secrete mucus, which traps bacteria, dust, and other particles.

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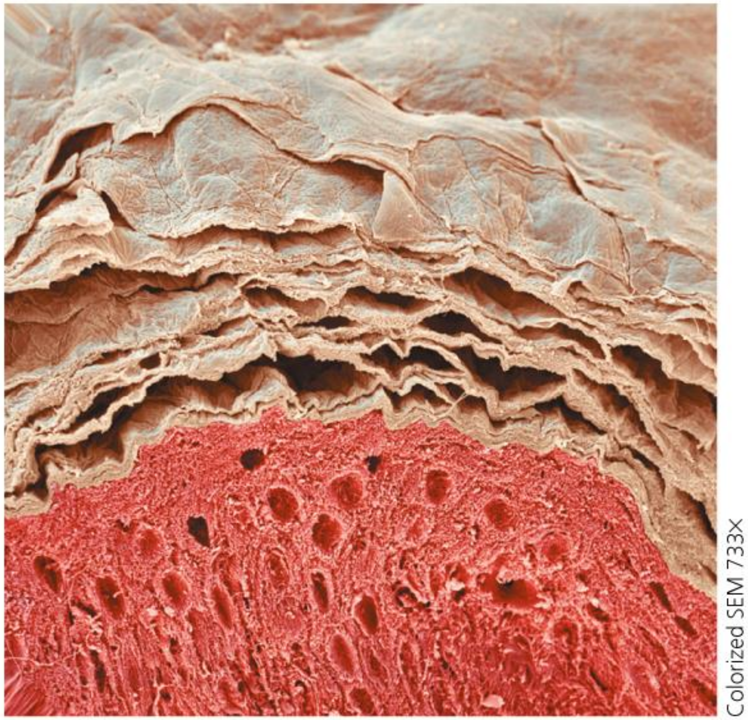
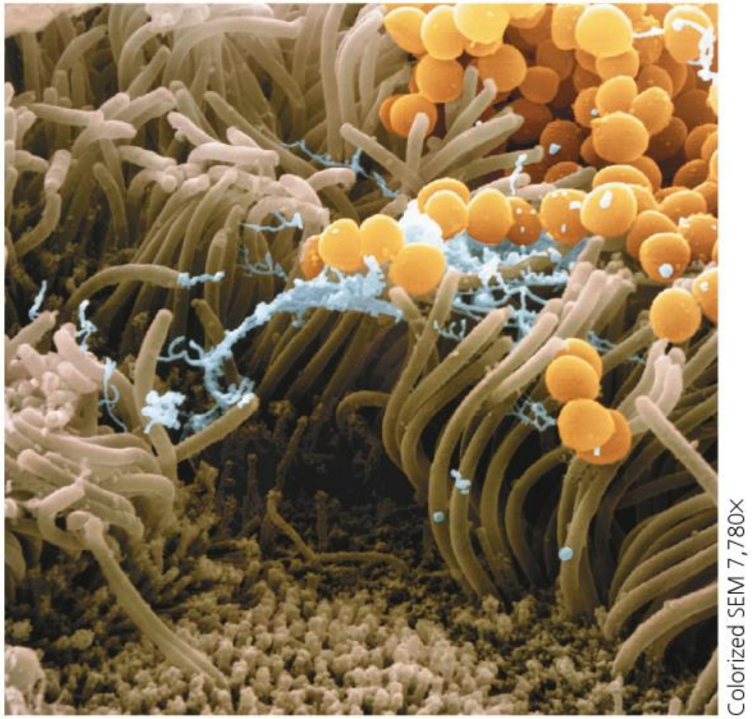
External innate defenses block or filter out pathogens



(a) Skin forms a protective barrier. The outermost layers of skin (brown) consist of dead cells that are continually shed and replaced by layers of living cells below (red).



(b) Cilia on cells in the nasal cavity. Mucus (blue) and trapped bacteria (yellow) are swept out of the body by hair-like cilia (brown).

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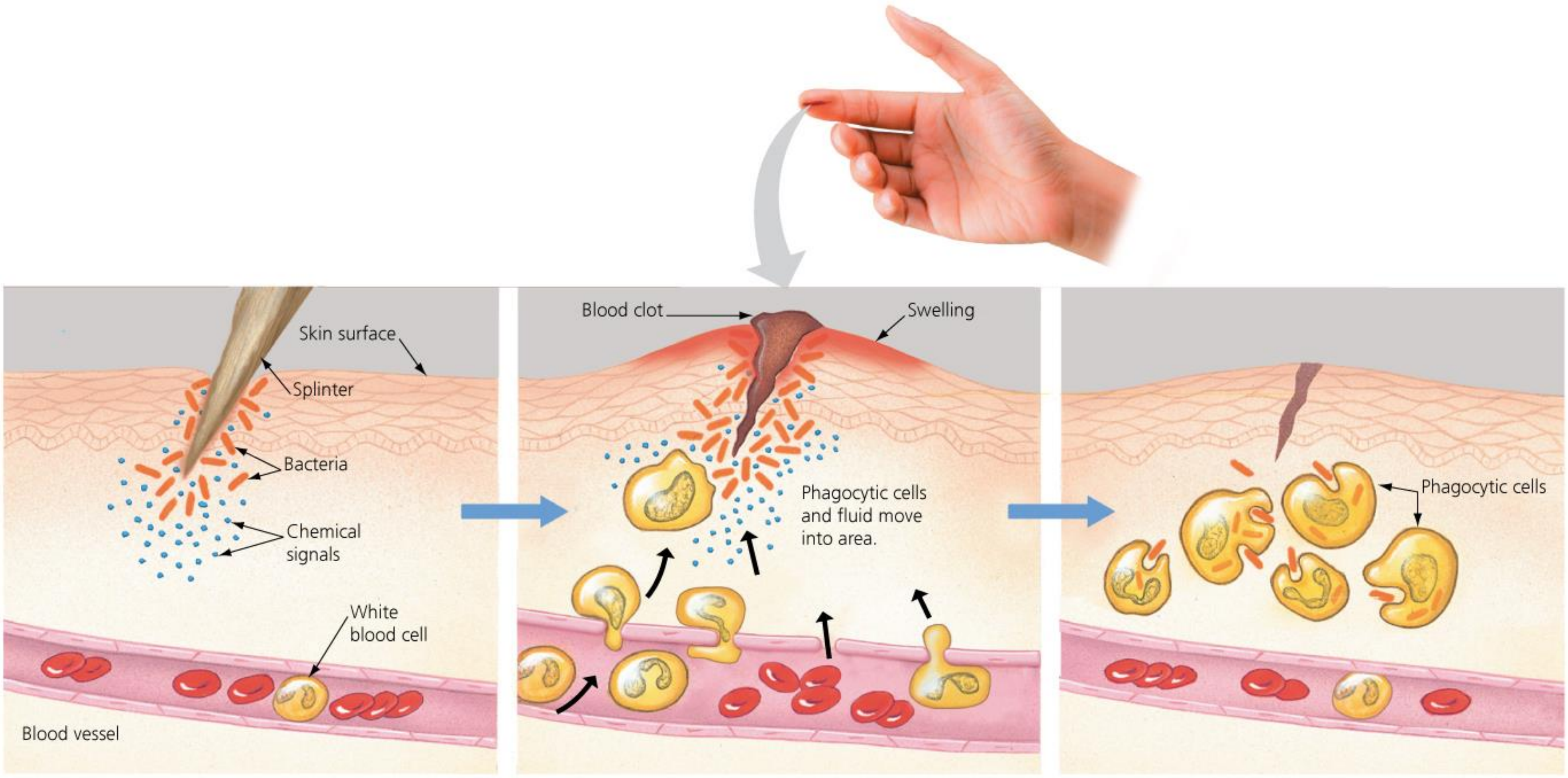
External Innate Defenses (2 of 2)

- External innate defenses also include chemical barriers in the form of antimicrobial secretions.
  - Sweat, saliva, and tears contain enzymes that disrupt bacterial cell walls.
  - The skin contains oils and acids that make it inhospitable to many microorganisms.
  - The cells of the stomach produce acid that kills most of the bacteria we swallow.

Internal Innate Defenses (1 of 2)

- The internal innate defenses depend on white blood cells and defensive proteins.
- Two types of white blood cells contribute to your internal innate defenses.
  1. **Phagocytic cells** (also called phagocytes) engulf foreign cells or molecules and debris from dead cells by phagocytosis, or "cellular eating."
  2. **Natural killer (NK) cells** recognize virus-infected and cancerous body cells.
- **Figure 24.3** shows the **inflammatory response**.

The inflammatory response



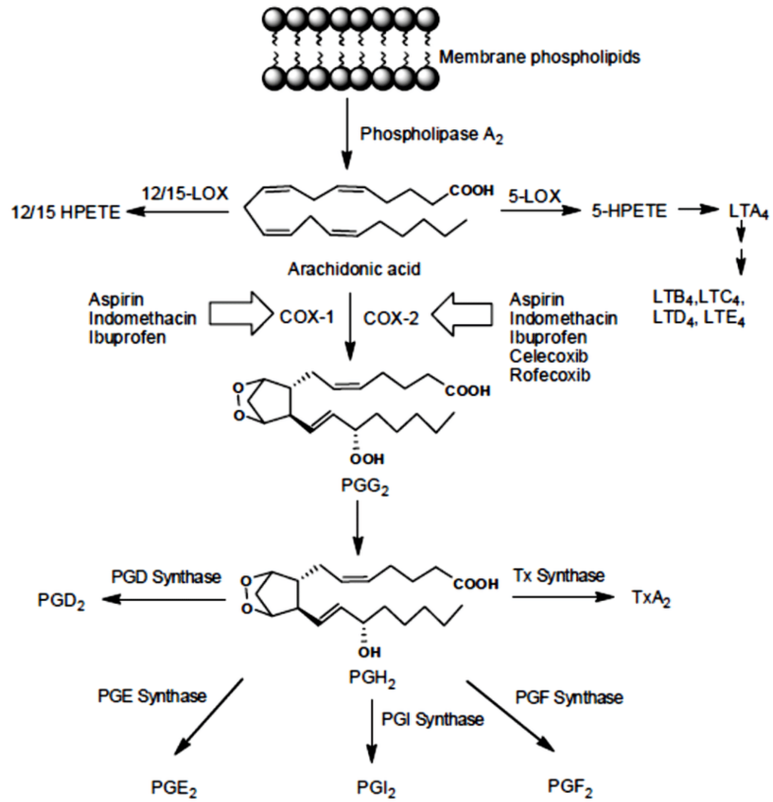
1 Tissue injury; release of chemical signals such as histamine

2 Dilation and increased leakiness of local blood vessels; migration of phagocytic cells to the area

3 Phagocytic cells engulf bacteria and cell debris; tissue heals

Internal Innate Defenses (2 of 2)

- Damaged cells release chemical signals that increase blood flow to the damaged area, causing the wound to turn red and warm.
- Anti-inflammatory drugs, such as ibuprofen, dampen the normal inflammatory response and help reduce swelling and fever.

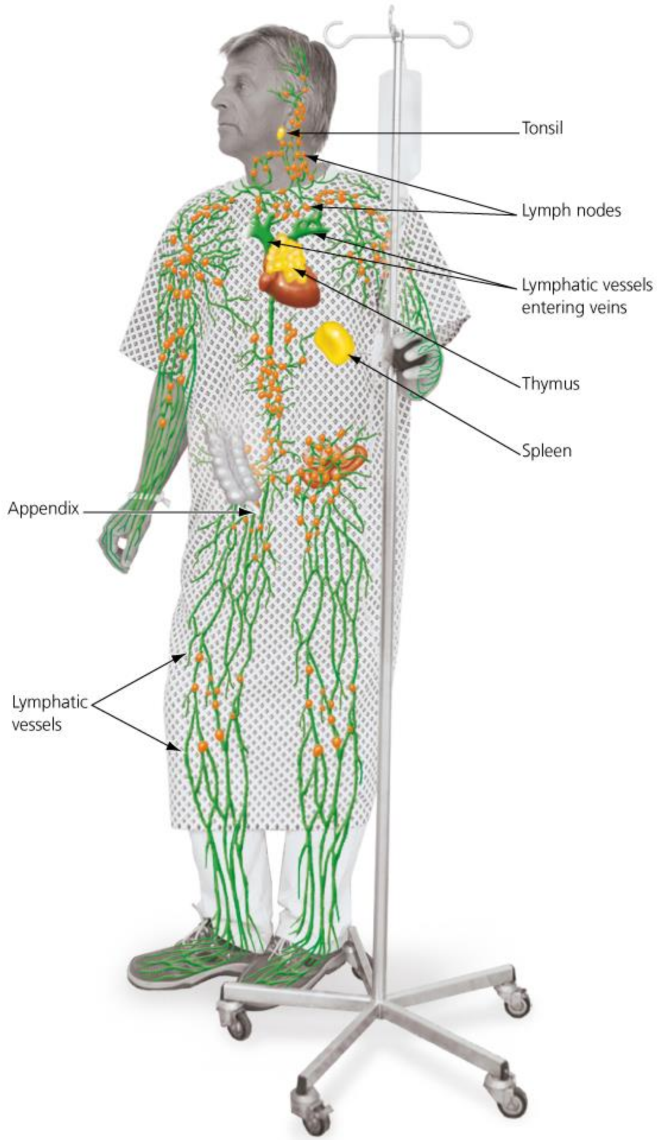


The Lymphatic System

- The lymphatic system includes a branching network of vessels, numerous **lymph nodes**, and the spleen, appendix, and tonsils.
  - The lymphatic vessels carry fluid called **lymph**, similar to the interstitial fluid surrounding body cells.
  - The two main functions of the lymphatic system are to return tissue fluid to the circulatory system and to fight infection.
  - To provide the best protection from pathogens, the body's immune system relies heavily on interactions of innate and adaptive immunities within the lymphatic system.

Spleen: 비장
Appendix: 충수
Tonsil: 편도선

The organs and vessels of the lymphatic system



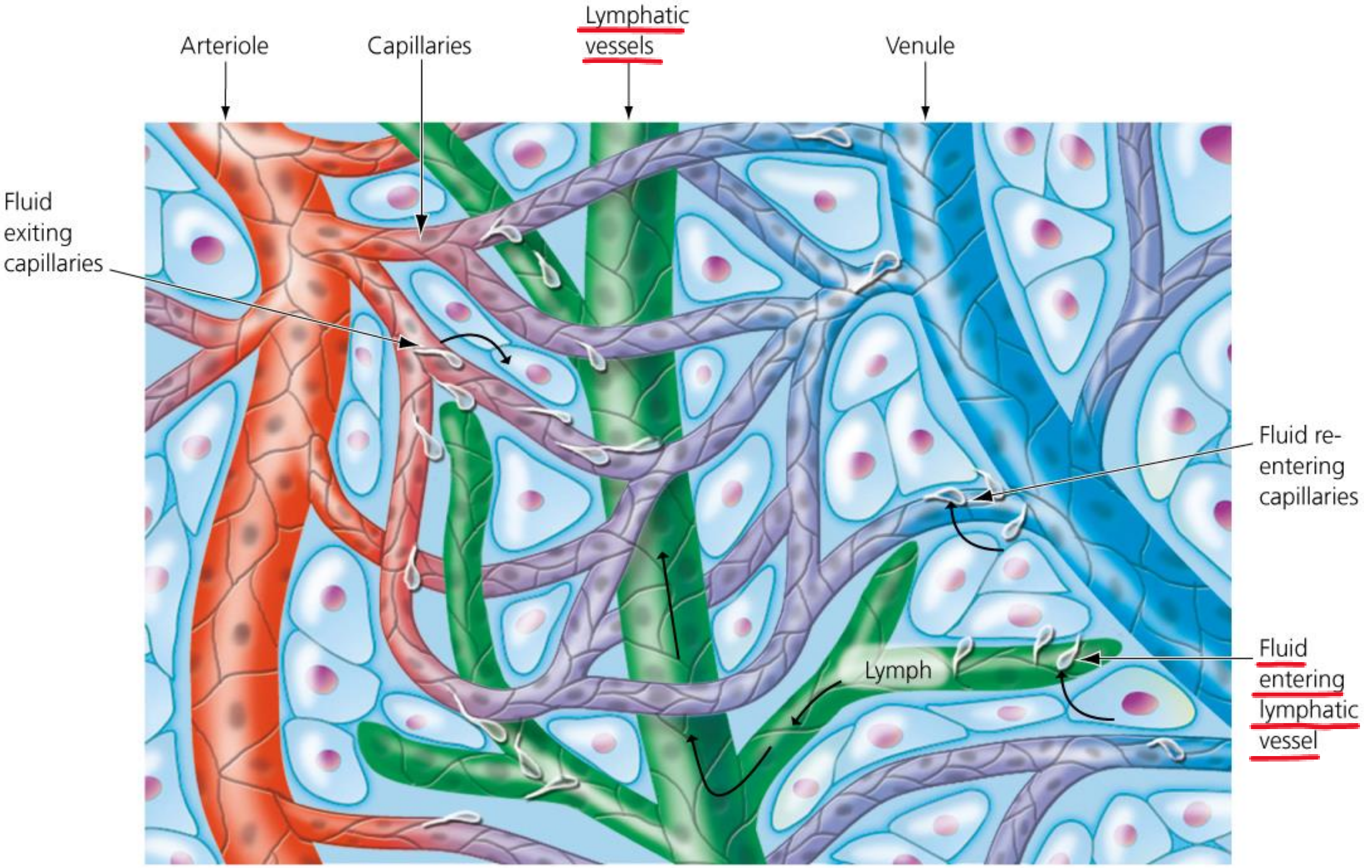
(a) The organs and vessels of the lymphatic system.

Thymus: 흉선, 가슴샘

Circulatory Function

- As blood travels through the circulatory system, fluid exits the blood through small gaps between the cells of the capillaries. This fluid then enters the interstitial space surrounding tissues.
  - Nutrients and wastes for all cells are exchanged in this interstitial fluid.
  - Most of this fluid reenters the blood and the circulatory system through capillaries, but some remains in the tissue.
  - This excess fluid, now called *lymph*, flows into small lymphatic vessels and then drains into larger lymphatic vessels.

Lymphatic vessels



(b) Lymphatic vessels. Interstitial fluid that enters lymphatic vessels is called lymph.

Immune Function

- Because lymphatic vessels penetrate nearly every tissue, lymph can pick up pathogens from infection sites just about anywhere in the body.
  - As this fluid circulates, phagocytic cells inside lymphatic tissues and organs engulf the invaders.
  - Lymph nodes are key sites where particular white blood cells called **lymphocytes** multiply during times of infection.

Adaptive Immunity (1 of 2)

- Adaptive immunity depends on two types of **lymphocytes** that recognize and respond to specific invading pathogens.
  - Like all blood cells, lymphocytes originate from stem cells in the bone marrow.
  - **B cells** fully develop and become specialized in the bone marrow.
  - Immature **T cells** migrate via the blood to the thymus, where they mature and become specialized.
  - B cells and T cells make their way to lymph nodes and other lymphatic organs and wait to encounter an invader.

Adaptive Immunity (2 of 2)

- Any molecule that elicits a response from a lymphocyte is called an **antigen**.
  - Most antigens are molecules on the surfaces of viruses or foreign cells, such as cells of bacteria or parasitic worms.
  - Any given pathogen will have many different antigens on its surface.
  - Antigens also include toxins secreted from bacteria, molecules from mold spores, pollen, and house dust, and molecules on cell surfaces of transplanted tissue.

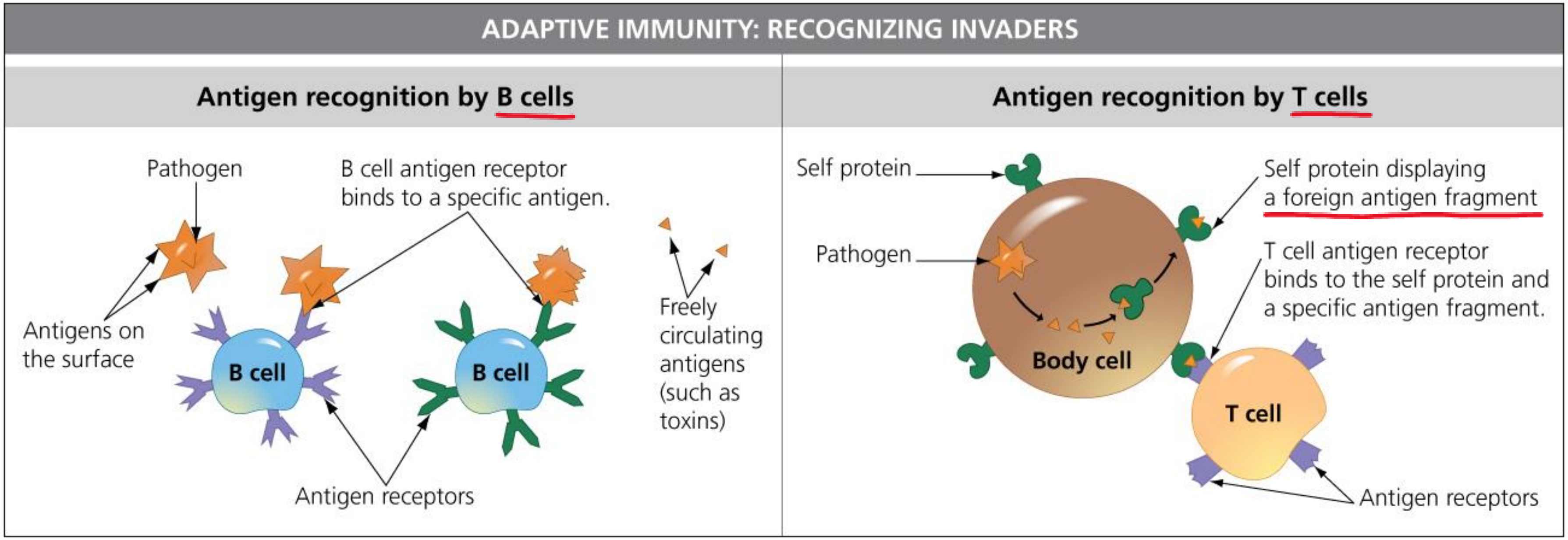
Step 1: Recognizing the Invaders (1 of 3)

- Unlike innate immunity defenses, which are ready to fight pathogens at any time, the adaptive immunity defenses, specifically B cells and T cells, must be primed before they attack foreign molecules.
  - Protruding from the surface of B cells and T cells are receptor proteins that bind to an antigen.
  - Each cell has about 100,000 copies of an antigen receptor that detects only a single type of antigen.
  - One cell may recognize an antigen on the mumps virus, for instance, whereas another detects an antigen on a tetanus-causing bacterium.

Step 1: Recognizing the Invaders (2 of 3)

- The antigen receptors of B cells and T cells have similarities in their structures, but they recognize antigens in different ways.
  - Antigen receptors on B cells specialize in recognizing intact antigens that are on the surface of pathogens or circulating freely in body fluids.
  - Antigen receptors on T cells only recognize fragments of antigens, and the fragments must be displayed, or presented on the surface of body cells by special proteins before T cells are activated.
  - **Figure 24.5** compares these two ways that antigens are recognized by the adaptive defenses.

Two ways antigens are recognized by the adaptive defenses



MHC

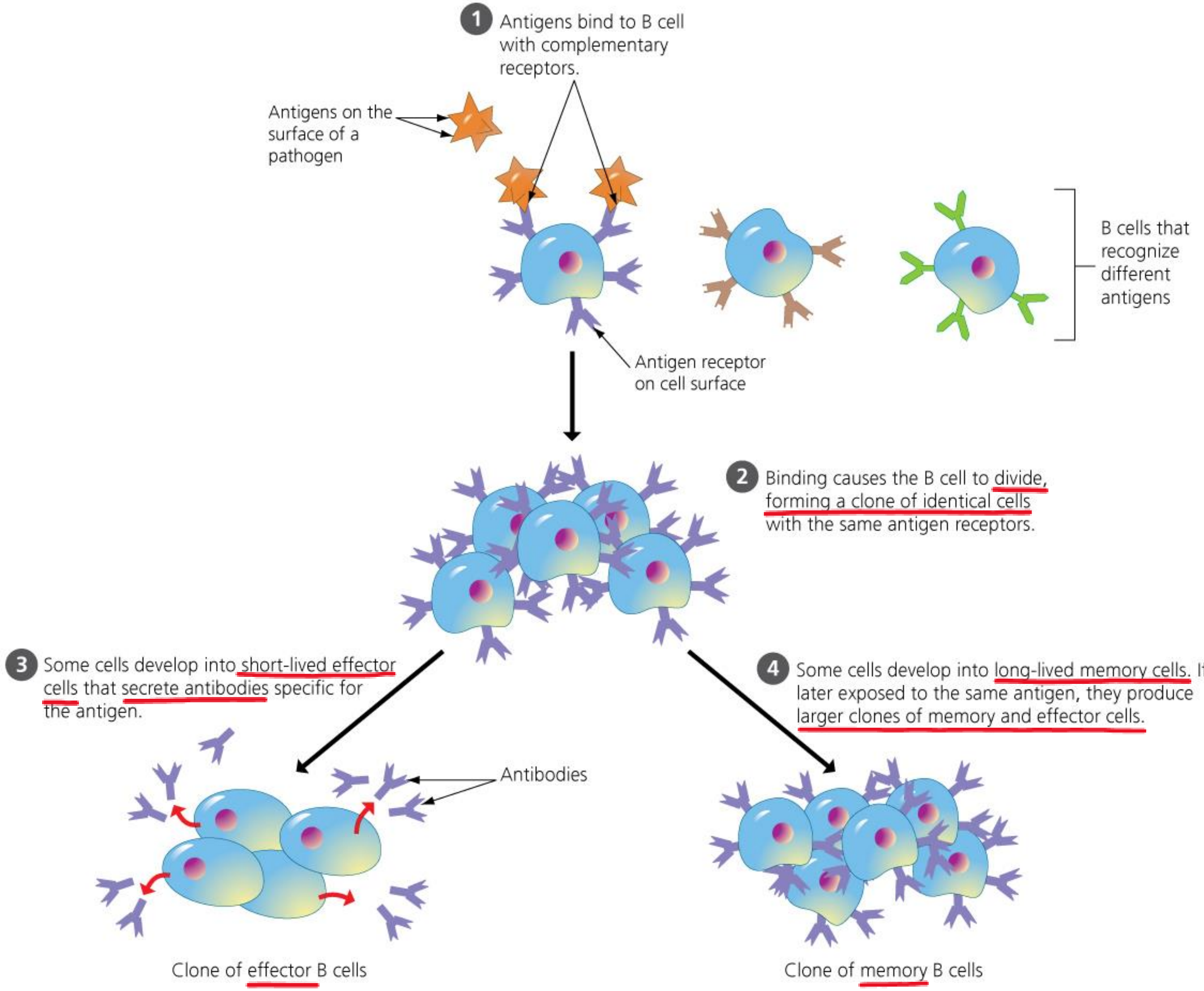
Step 1: Recognizing the Invaders (3 of 3)

- The immune system develops a great diversity of B cells and T cells—enough to recognize and bind to just about every possible antigen.
  - A small population of each kind of lymphocyte lies in wait in your body, ready to recognize and respond to a specific antigen.
  - Over your lifetime, only a tiny fraction of your B cells and T cells will ever be used, but all the variations are available if needed.
  - It is as if the immune system maintains a huge standing army of soldiers, each made to recognize one particular kind of invader.

Step 2: Cloning the Responders

- How does the body marshal enough of the right kind of lymphocyte to fight a specific invading pathogen?
  - The key is a process called **clonal selection**.
  - **Figure 24.6** illustrates how clonal selection of lymphocytes works, using B cells as an example.

Clonal selection during the first exposure to an antigen



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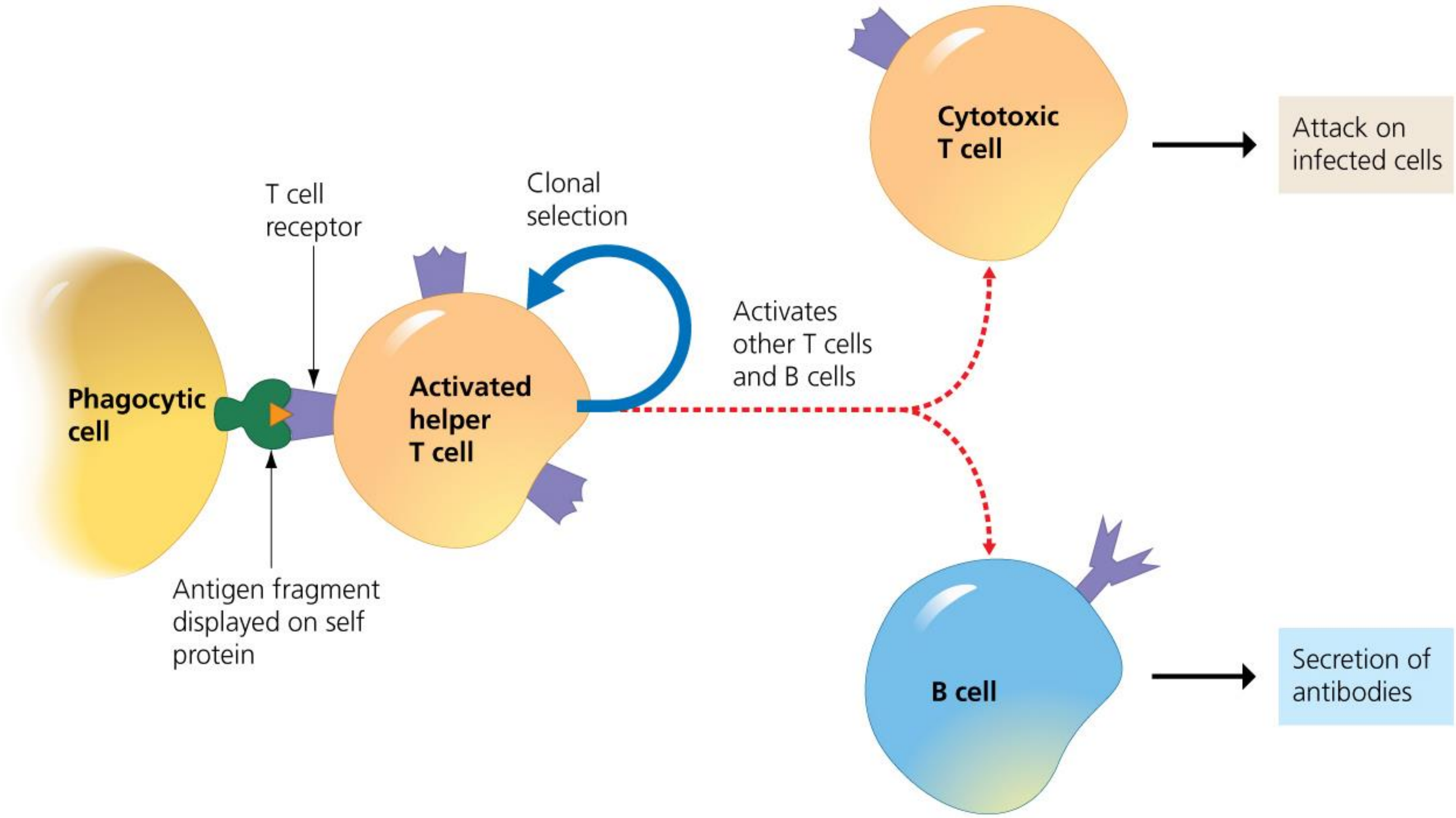
Step 3: Responding to Invaders: The Helper T Cell Response

- **Helper T cells** do not directly carry out attacks on pathogens but aid in stimulating both the B cells and other T cells in their responses.
  - Each helper T cell only recognizes a specific antigen fragment displayed on self proteins. Helper T cells are activated when particular white blood cell types "advertise" the antigen.
  - Once activated, helper T cells give rise to effector helper T cells and memory helper T cells through clonal selection. As seen in **Figure 24.7**, effector helper T cells respond to infection by stimulating the activity of B cells and cytotoxic T cells.

Helper T cells: effector helper T cells, memory helper T cells
Antigen presenting cells
B cells, cytotoxic T cells

The roles of an activated helper T cell



The Helper T Cell Response

- Because the helper T cell has a central role in adaptive immunity, the destruction of this cell type has devastating consequences. The human immunodeficiency virus (HIV) infects helper T cells.
  - HIV infection causes helper T cell numbers to decline significantly.
  - If not treated, HIV infection results in acquired immune deficiency syndrome (AIDS).
  - Individuals with AIDS lack a completely functional immune system and die from exposure to other infectious agents.

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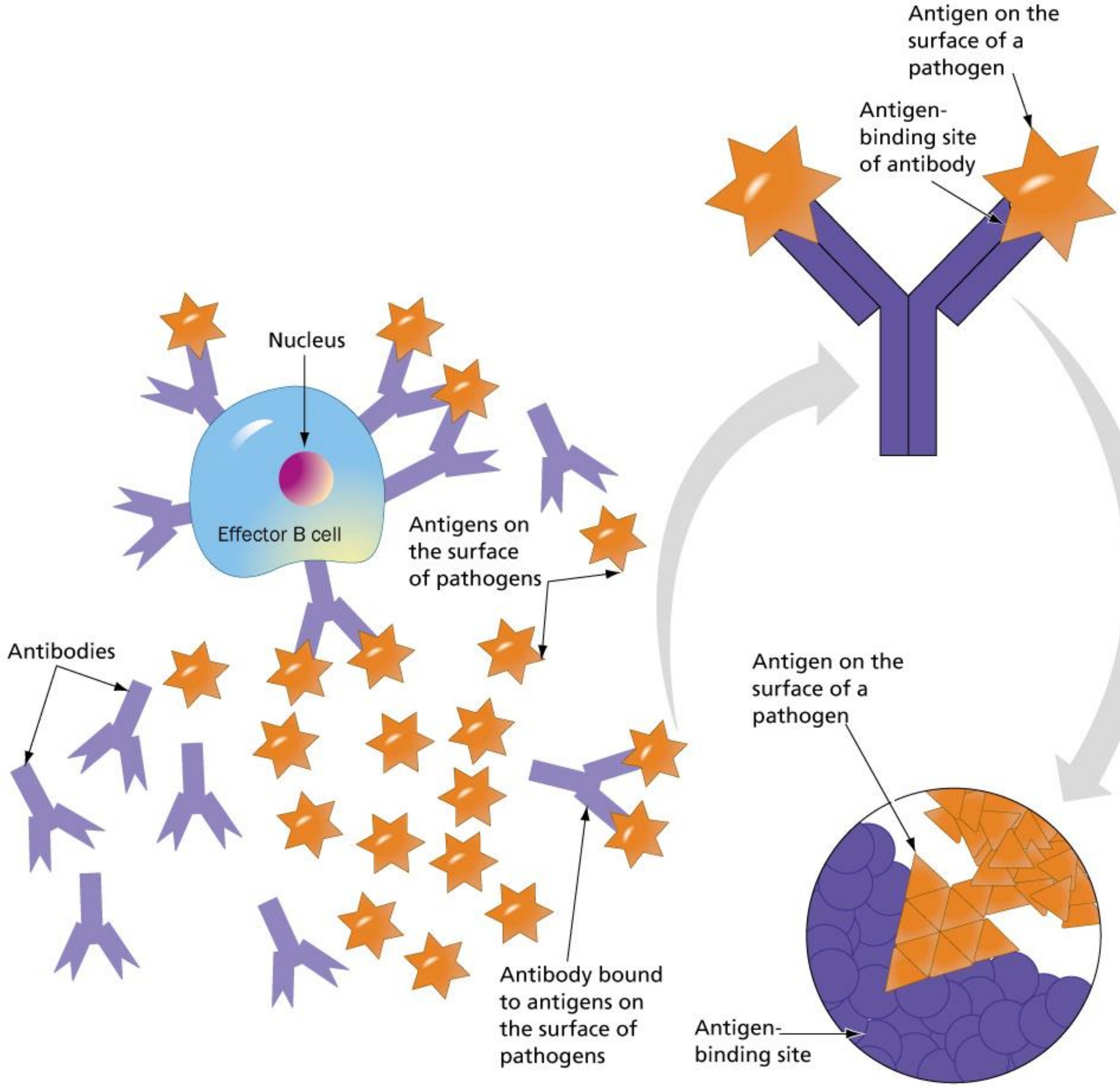
The B Cell Response (1 of 4)

- By secreting antibodies into the blood and lymph, B cells defend primarily against pathogens circulating in body fluids.
  - The information for making a specific antibody is coded within the DNA in the nucleus of a B cell.
  - Four polypeptides are synthesized and joined to form a single Y-shaped antibody protein that is then secreted. The tip of each “Y” forms a region, an antigen-binding site, that will recognize and bind to a specific antigen in a lock-and-key structure.
  - One effector B cell can produce up to 2,000 antibodies per second.

The B cell response (2 of 4)



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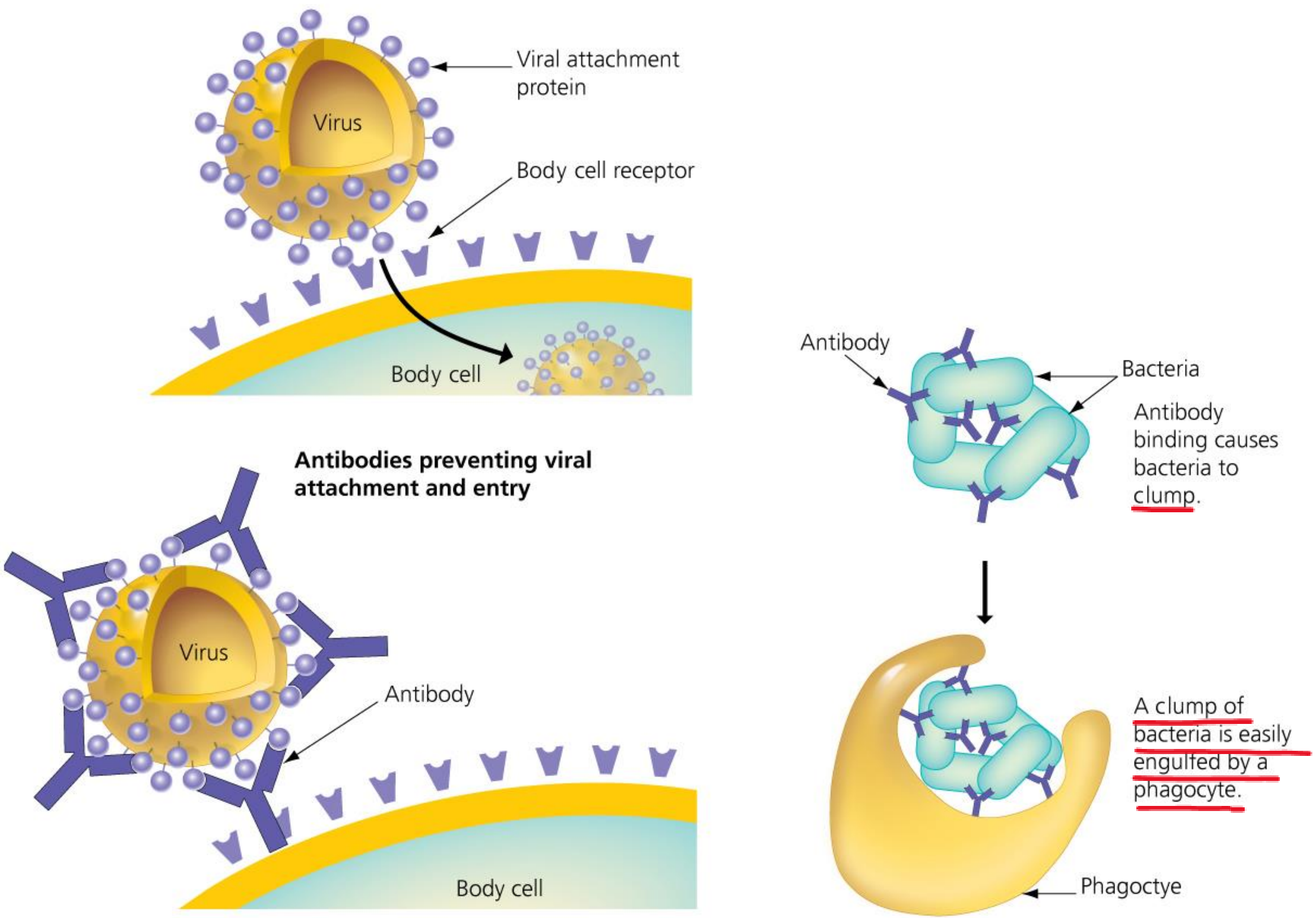
The B Cell Response (3 of 4)

- In another example of the relationship between structure and function in biological systems, the shape of an antibody is related to its ability to bind to and eliminate antigens.
- As shown in **Figure 24.9**, the binding of antibodies to antigens blocks or helps destroy an invader.

Binding of antibodies to antigens for the destruction of an invader



(a) Antibodies block a virus from entering a body cell.

(b) Antibodies enhance phagocytosis.

The B Cell Response (4 of 4)

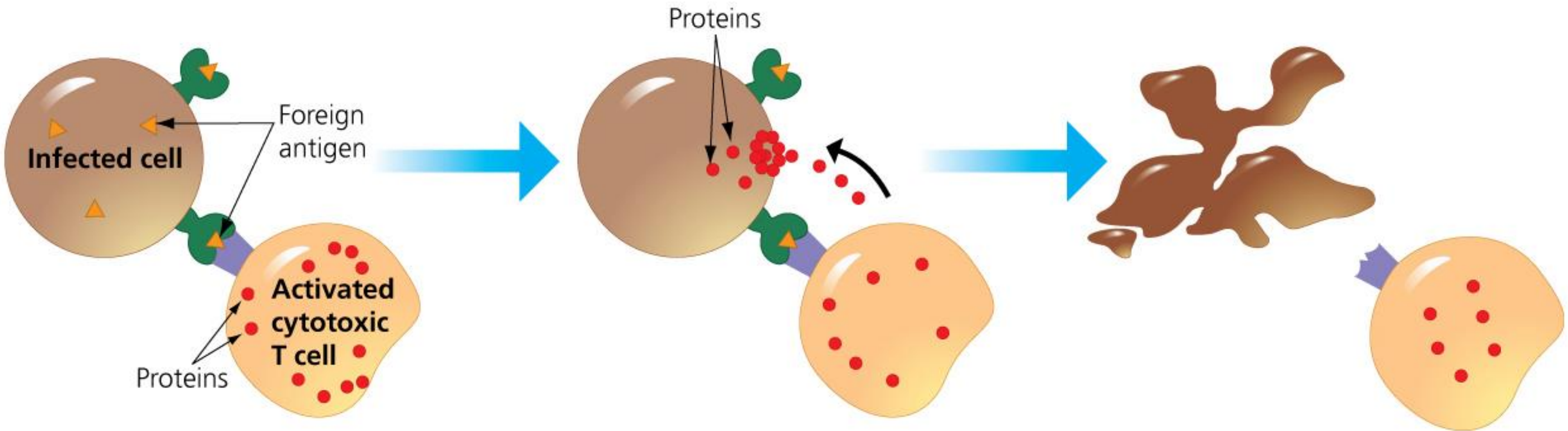
- Because of their ability to tag specific molecules or cells, antibodies are useful in medicine.
  - Herceptin, a genetically engineered antibody, treats certain cases of aggressive breast cancer.
  - In home pregnancy tests, antibodies are used to detect a hormone called human chorionic gonadotropin (HCG), present in the urine of pregnant women. When a testing strip is dipped into urine, antibodies on the strip bind to HCG, causing the strip to change color.

Her2: human epidermal growth factor receptor 2
hCG: 인간 융모성 성선자극호르몬

The Cytotoxic T Cell Response (1 of 3)

- **Cytotoxic T cells** defend against pathogens that have entered body cells.
  - Cytotoxic T cells are the only T cells that actually kill infected cells. They identify infected body cells because foreign antigen fragments are “advertised,” or bound to a self protein.
  - **Figure 24.10** shows the steps of the cytotoxic T cell response.

The cytotoxic T cell response (2 of 3)



1 Cytotoxic T cell binds to infected cell, becoming activated.

2 Proteins that trigger cell death enter the infected cell.

3 Infected cell dies.

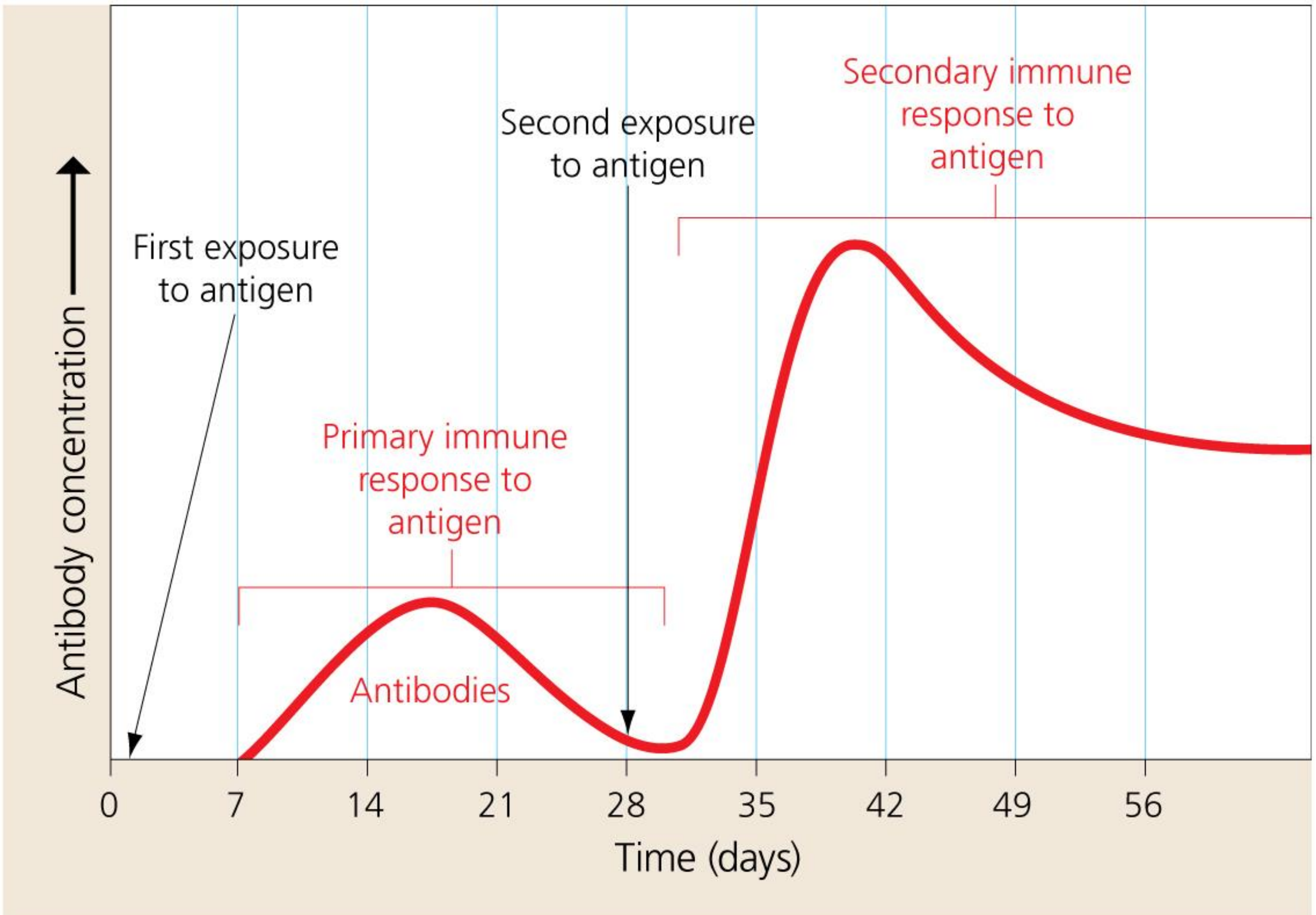
The Cytotoxic T Cell Response (3 of 3)

- The cytotoxic T cell response does not always work in a person's favor.
  - When an organ is transplanted from a donor into a recipient, the newly transplanted cells contain self proteins which do not match those on the recipient's cells. Thus, the recipient's cytotoxic T cells tag the transplanted cells as foreign and kill them, ultimately causing organ rejection.
  - To minimize this problem, doctors look for a donor with self proteins that match the recipient's as closely as possible, and they administer drugs to suppress the recipient's immune response.

Step 4: Remembering Invaders (1 of 3)

- Memory cells can last decades in the lymph nodes, ready to be activated if the antigen reappears.
  - When an antigen is encountered a second time, the secondary immune response will be more rapid, of greater magnitude, and of longer duration.
  - Memory cells are thus a good example of how information can be stored and used by living systems—in this case, information about previously encountered pathogens helps the body quickly defeat them during future exposures.
  - **Figure 24.11** shows antibody production during the two phases of the memory B cell response.

Antibody production during the two phases of the B cell response



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Step 4: Remembering Invaders (2 of 3)

- Immunity is obtained after an infection, but it can also be achieved artificially by **vaccination** (also called immunization), in which the immune system is confronted with a **vaccine** composed of an inactive or otherwise harmless version of a pathogen.
  - Vaccination induces the primary immune response that produces memory cells.
  - When a person has been successfully vaccinated, the immune system responds quickly and effectively to the pathogen. Such protection may last for life.

Step 4: Remembering Invaders (3 of 3)

- In the United States, most children receive a series of vaccine shots soon after birth, protecting against
  - diphtheria/pertussis/tetanus (DPT),
  - polio, hepatitis, chicken pox, and
  - measles/mumps/rubella (MMR).
- Widespread childhood vaccination in the United States has virtually eliminated several of these diseases, including polio and mumps. The World Health Organization of the United Nations confidently declared smallpox to be completely eradicated in 1980.

Diphtheria: 디프테리아
Pertussis: 백일해
Tetanus: 파상풍
Polio: 소아마비
Chicken pox: 수두
Mumps: 볼거리
Rubella: 풍진, 독일 홍역
Smallpox: 천연두

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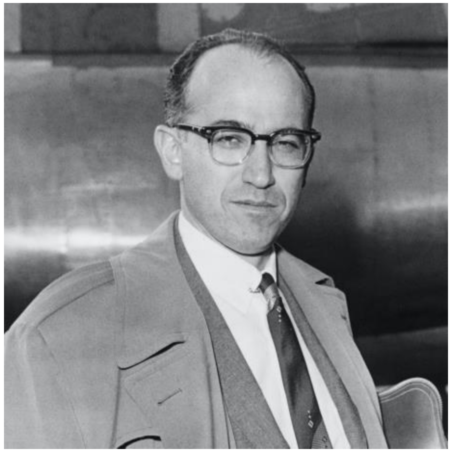
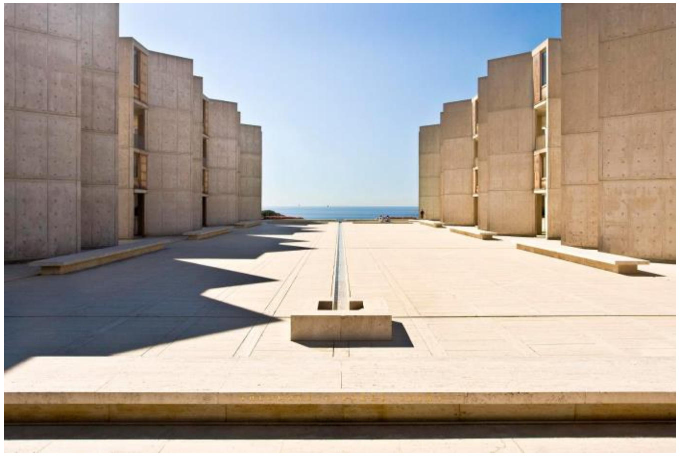
The Process of Science: How Do We Know Vaccines Work? (1 of 4)

- **Background:** Polio is a contagious disease that attacks the nervous system, causing temporary or permanent paralysis in the muscles that control walking, swallowing, and breathing. In 1954, Dr. Jonas Salk developed a poliovirus vaccine based on two prior medical observations.
    1. People infected with a disease were often protected from getting that same disease again.
    2. Infectious diseases are caused by microbes.
- Based on these observations, Salk set out to develop a vaccine to prevent polio.





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The Process of Science: How Do We Know Vaccines Work? (2 of 4)

- **Method:** Dr. Salk hypothesized that an inactivated form of the polio virus could prime the immune system. To test his vaccine, Salk conducted a controlled experiment.
- Because paralysis from the disease is more likely to occur in children than in adults, approximately 400,000 children were enrolled in a trial.
  - Children in the control group were injected with salt water (a placebo).
  - Children in the experimental group were injected with the polio vaccine.

The Process of Science: How Do We Know Vaccines Work? (3 of 4)

- The experiment was "double-blind," meaning that none of the participants nor the researchers knew to which group each child was assigned.
- Once all the children were injected, scientists waited to see how many of the children became infected with the poliovirus over a six-month period.

The Process of Science: How Do We Know Vaccines Work? (4 of 4)

- **Results**: The results, published in 1955, were clear: There were significantly fewer cases of polio and of its paralytic symptoms in children who received the vaccine injection than in children who received the placebo.
  - Salk's polio vaccine was effective. Within a few years, the number of polio cases in the United States dropped by 85–90%.
  - By 1979, polio had been eradicated in the United States. Interestingly, Salk chose not to patent his vaccine, giving up any personal profits. When asked about this, Salk asked, "Could you patent the sun?"

Development of the polio vaccine

(a) A 1950s polio ward in Los Angeles housing dozens of children in iron lungs

(b) Jonas Salk (1914–1995, on right) is about to inject a young girl with the polio vaccine that he developed.

Results from the Salk Polio Vaccine Trials of 1954			
Study Group	Number of children	Number of polio cases	Number of polio cases involving paralysis
Control group (received placebo injection)	201,229	142	115
Experimental group (received polio vaccine)	200,745	57	33

(c) Data from Salk's 1954 polio vaccine trial

Immune Disorders - Allergies

- If the intricate interplay between innate immunity and adaptive immunity goes awry, problems can arise that range from mild irritations to deadly diseases.
- An **allergy** is an exaggerated sensitivity to an otherwise harmless antigen in the environment.
  - Antigens that cause allergies are called **allergens**.
  - Common allergens include protein molecules on pollen grains, on the feces of tiny mites that live in house dust, in animal dander (shed skin cells), and in various foods.

Mite: 진드기
Dander: 털의 비듬

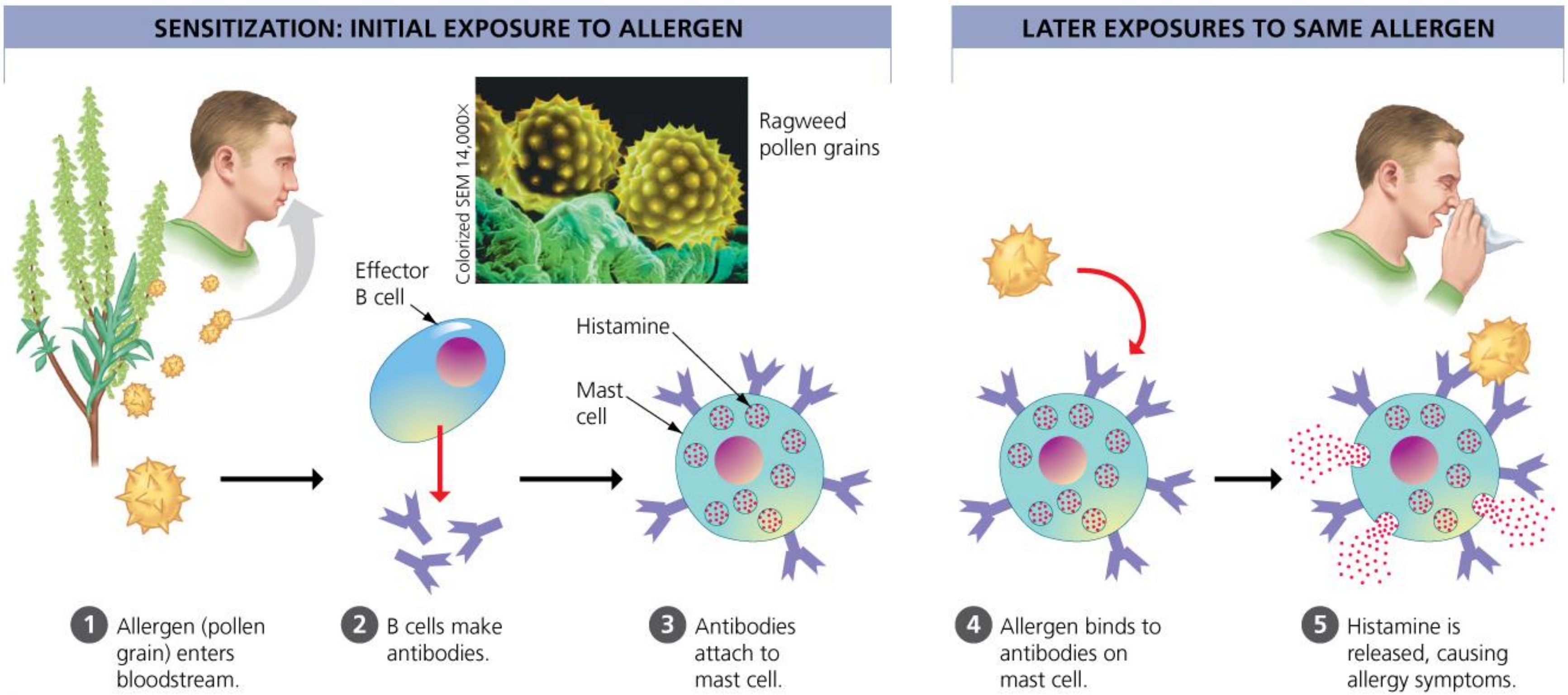
The lone star tick

Tick: 진드기

Allergies (1 of 2)

- New allergens are periodically discovered. For example, an allergy to meat develops after a tick bite, common in southern and eastern states.
- The symptoms of an allergy result from a two-stage reaction outlined in **Figure 24.14**.

How allergies develop



Allergies (2 of 2)

- Allergic reactions range from seasonal nuisances to severe, life-threatening responses.
  - Some people are extremely sensitive to certain allergens, such as the venom from a bee sting or allergens in peanuts or shellfish. Any contact with these allergens causes a sudden release of inflammatory chemicals; blood vessels dilate abruptly, causing a rapid and potentially fatal drop in blood pressure, a condition called shock.
  - Anaphylactic shock is an especially dangerous type of allergic reaction but can be counteracted with injections of the hormone epinephrine.

Shellfish: 조개
Anaphylactic shock: 과민성 쇼크

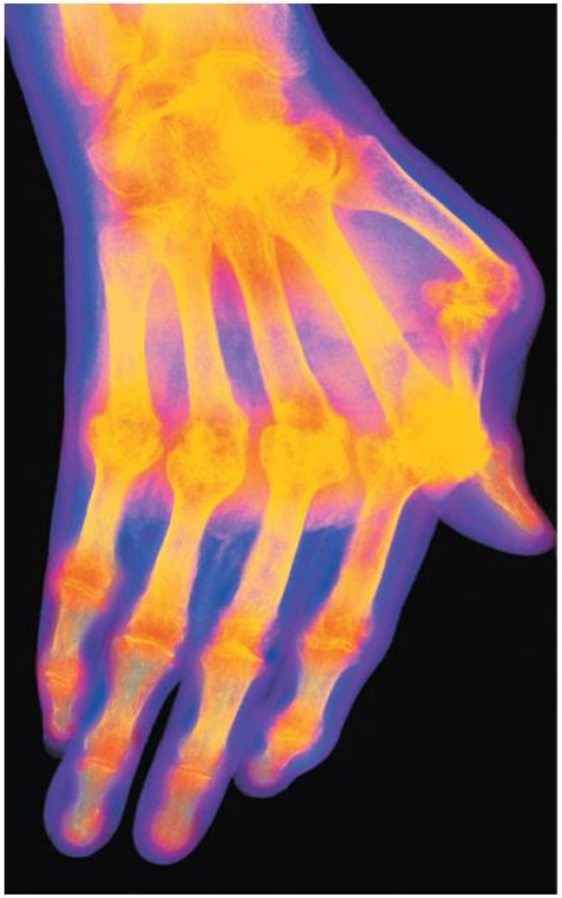
Single-use epinephrine syringes

Autoimmune Diseases

- Your immune system's ability to identify its own cells enables your body to battle foreign invaders without harm to itself.
  - In **autoimmune diseases**, the immune system actually turns <u>against the body's own molecules</u>.
  - Examples include systemic <u>lupus erythematosus (lupus)</u>, <u>type 1 (insulin-dependent) diabetes</u>, <u>multiple sclerosis</u>, <u>rheumatoid arthritis</u>, and <u>Crohn's disease</u>, a chronic inflammation of the digestive tract, which may be caused by an <u>autoimmune reaction against bacteria that normally inhabit the intestinal tract</u>.

Lupus: 낭창 (결핵성 피부병의 일종)
Multiple sclerosis: 다발성 경화증
Rheumatoid arthritis: 류마티스성 관절염

Rheumatoid arthritis



Immunodeficiency Diseases

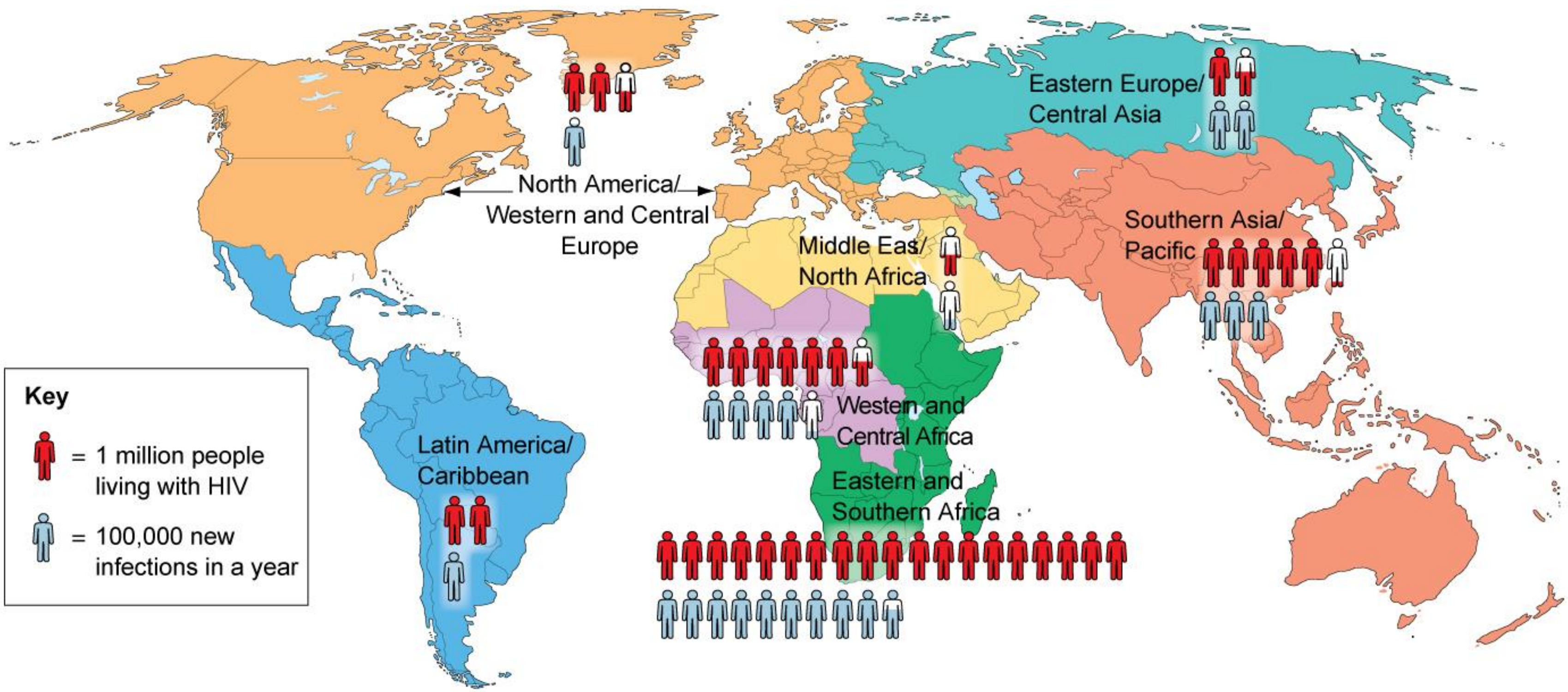
- **Immunodeficiency diseases** result when one or more of the components of the immune system are lacking and leave affected people more susceptible to infections.
- Immunodeficiencies may arise
  - through inborn conditions such as severe combined immunodeficiency (SCID),
  - from acquired illness such as Hodgkin's disease, a type of cancer that affects lymphocytes, or
  - from radiation therapy or drug treatments used against many cancers.

Severe combined immunodeficiency: 중증 복합형 면역 부전증
Hodgkin's disease: 호지킨병 (악성 림프종)

AIDS (1 of 2)

- Since the epidemic was first recognized in 1981, AIDS has become a worldwide scourge.
  - AIDS is caused by HIV, a virus with some special properties.
  - HIV is deadly because it destroys the immune system, leaving the body defenseless against most invaders. HIV can infect a variety of cells, but it most often attacks helper T cells—the cells that activate cytotoxic T cells and B cells.
  - If not treated with anti-HIV drugs, a final stage of HIV infection occurs, known as AIDS.

Statistics on HIV infection around the world in 2015



Data from: *AIDS by the numbers*, UNAIDS Joint United Nations Programme on HIV/AIDS, Geneva, Switzerland (2016), unaids.org. All data is from 2015.

AIDS (2 of 2)

- Drugs, vaccines, and education are areas of focus for prevention of HIV infection.
  - Drug development has led to a drastic reduction in transmission rates of HIV from mother to child.
  - In 2012, the Federal Drug Administration (FDA) approved the first HIV prevention pill for people who have a high risk of infection.
  - A vaccine against HIV has been challenging to develop, and one does not yet exist.
  - The most effective HIV/AIDS prevention is education. Safe sex behaviors, such as abstinence and using condoms, can save many lives.

Evolution Connection: Viral Evolution versus the Flu Vaccine (1 of 5)

- To understand how vaccines work or fail, we need to keep in mind biology's unifying theme: evolution.
  - As viruses reproduce, mutations occur, generating new strains of a virus.
  - These new strains are different enough that they can cause disease in individuals who had immunity to the ancestral strain.

Evolution Connection: Viral Evolution versus the Flu Vaccine (2 of 5)

- The flu virus is associated with 3,000–49,000 deaths in the United States in a given year.
  - Most vaccines provide protection for several years or even a lifetime. Why, then, do we need a flu vaccine every year?
  - The flu virus mutates rapidly, so the antigens from one year’s virus often change significantly by the next year. Our immune system doesn’t have memory cells that will protect us from flu strains that are circulating currently or that will in future years.

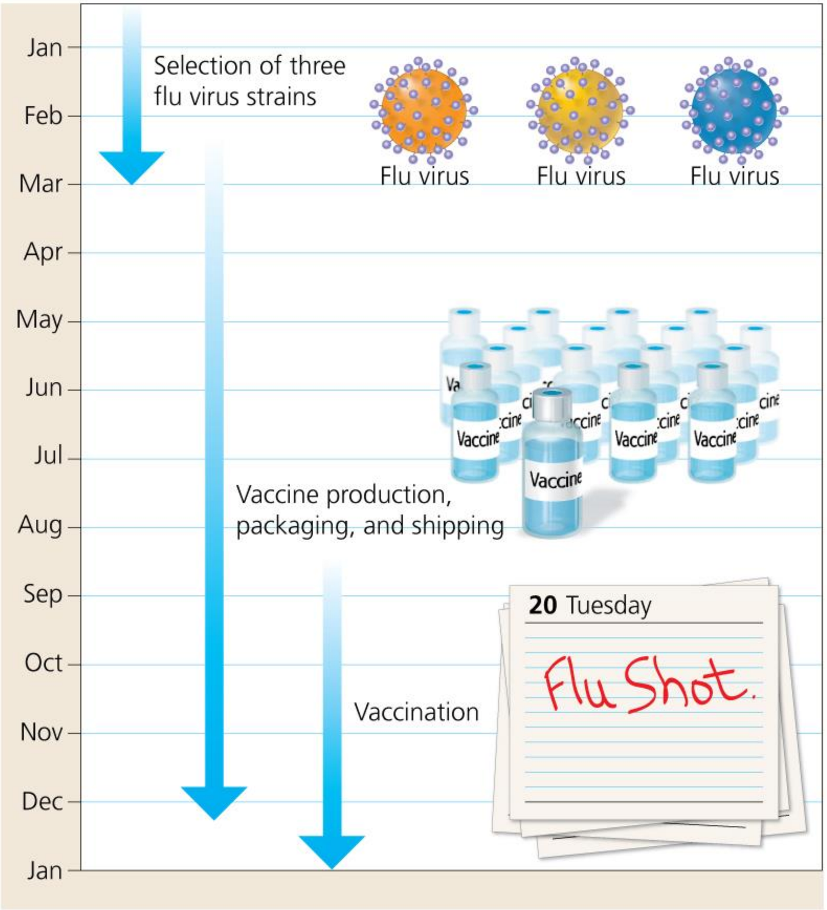
Evolution Connection: Viral Evolution versus the Flu Vaccine (3 of 5)

- Developing a flu vaccine is a challenge.
  - Several major flu virus strains circulate in a given flu season, each strain mutating at a high rate.
  - To examine which strains are likely to cause the next flu season's epidemic, scientists use a world-wide surveillance system, a computer program that gathers data on flu outbreaks from countries around the globe.

Evolution Connection: Viral Evolution versus the Flu Vaccine (4 of 5)

- From these data, scientists typically select the three most prevalent strains for use in a single vaccine. The timing of this choice is tricky.
- The selection of strains occurs between January and March, but it takes more than six months to produce and deliver a vaccine to healthcare providers.
- In the interim, viral evolution continues, and the developed flu vaccine becomes less and less of a match to the flu virus strains that ultimately circulate. Thus, the vaccine given to us in November is an “educated guess,” based on a strain that circulated several months earlier.

The flu vaccine production time line



Pearson

Evolution Connection: Viral Evolution versus the Flu Vaccine (5 of 5)

- Scientists are working toward what they are calling a "universal flu vaccine" that would provide broad protection against several flu virus strains and would not be needed each year.
- For now, however, the battle continues, with medical science on one side and the constantly evolving flu viruses on the other side.

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